

The High-Redshift Accretion Atlas

Project Status Through v4

Project status manuscript

22 August 2026

Abstract

JWST has identified accreting black-hole candidates at epochs where the available growth time is short and inferred physical quantities remain method-dependent. This project develops a reproducible, source-tracked, uncertainty-aware atlas that ranks observational leverage under explicit seed and accretion assumptions. Project v1 standardized 23 JADES broad-line AGN measurements at $z \geq 4$; v2 added point and Monte Carlo rankings plus figure prototypes; v3 added 37 Taylor et al. CEERS/RUBIES measurements; and v4 adds 20 Matthee et al. EIGER/FRESCO and 16 Lin et al. ASPIRE measurements. The current release has 96 measurements representing 94 physical objects. A generalized identity layer retains every literature measurement while preventing CEERS-2782/RUBIES-EGS-50052 and GS-204851/GOODS-S-13971 from being counted twice at object level. Under the baseline $z_{\text{seed}} = 30$, $\epsilon = 0.1$, and no-merger reference, the framework remains diagnostic: rankings express observational growth pressure and do not prove a unique seed or accretion channel. Next steps are manuscript consolidation, measurement-choice sensitivity, method-aware object classes, and duty-cycle diagnostics.

1 Scientific objective

The motivating question is simple: *how can the reported black holes become so massive so early?* The answer depends jointly on the initial seed mass, formation redshift, lifetime-average accretion rate, radiative efficiency, merger history, and the reliability of the inferred black-hole mass. Across the literature, these quantities are reported using different methods and assumptions. Consequently, the main project goal is not to draw one preferred growth track, but to construct an observational triage framework that separates:

- objects that remain growth-challenging under reasonable uncertainty and systematic tests;
- objects whose interpretation changes substantially with the inferred mass or growth assumptions; and
- measurements whose improvement would most change the physical conclusion.

The working thesis is that a standardized, provenance-aware atlas can identify a robust set of high-leverage objects for follow-up and detailed seed/accretion modeling. Formation scenarios are therefore treated as diagnostic coordinate systems, not as proof of a unique formation pathway.

2 Chronological development through v4

2.1 Stage 1: a controlled observational sample

The project begins with one relatively homogeneous class from one source paper: the JADES broad-line AGN catalogue of Juodžbalis et al. (2026) [2]. The source registry records the paper, tables used, selection,

survey, and inference methods. The raw table contains 34 measurements; the reproducible standardization pass applies the $z \geq 4$ selection and produces 23 measurement rows spanning $z = 4.133\text{--}8.913$. The current quality classification contains 18 robust and 5 tentative measurements. Four high-redshift candidates—GS-20057765, GS-20030333, GS-164055, and GN-4685—use single-epoch broad- $H\beta$ virial masses and are explicitly classified as stack-supported tentative detections because their individual broad-component detections are not formally significant. Source-adopted CIGALE host masses are retained when BEAGLE is unavailable or the host is significantly extended; this supplies host masses for GS-200679, GS-20030333, and GS-164055 that were previously omitted.

The processed catalogue preserves coordinates, redshift provenance, black-hole mass and asymmetric uncertainty, host stellar mass, bolometric luminosity, reported Eddington ratio, inference methods, source-table provenance, notes, interpretation tags, a controlled detection-evidence category, derived quality flags, and explicit optional-field missingness. It also compares each reported Eddington ratio with the value implied by the tabulated black-hole mass and bolometric luminosity. Missing host or lensing information is retained and flagged rather than silently imputed.

Table 1: Catalogue chronology and current output scale.

Product	Current size
Raw source measurements	34
Processed measurements at $z \geq 4$	23
Robust / tentative measurements	18 / 5
Taylor Table 1 measurements (all redshifts)	63
Taylor measurements / objects at $z \geq 4$	37 / 36
v3 measurements / physical objects at $z \geq 4$	60 / 59
Matthee / ASPIRE source measurements	20 / 16
v4 measurements / physical objects at $z \geq 4$	96 / 94
Complete scenario-evaluation rows	8,832
Required- f_{Edd} rows	2,208
Required-seed rows	2,208
Per-object spin/merger parameter maps	138
Per-object seed-redshift maps	23

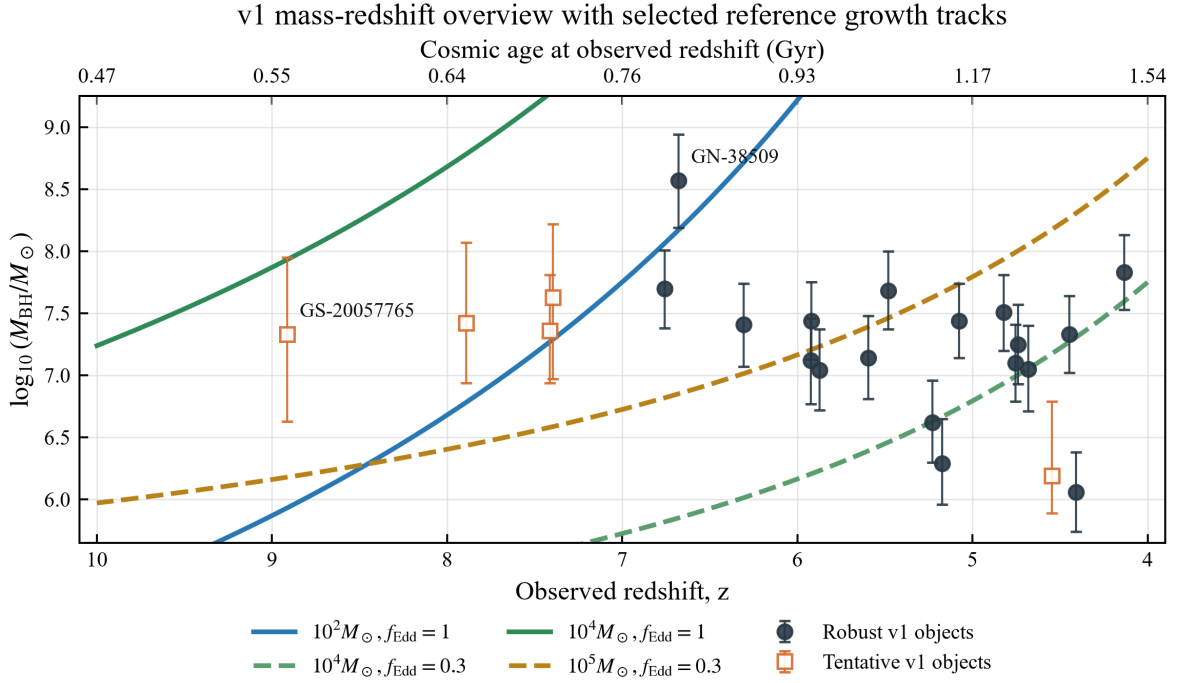
2.2 Stage 2: a reproducible physical model

The growth model follows the exponential form used by Dayal (2024) [1],

$$M_{\text{BH}}(t_{\text{obs}}) = M_{\text{seed}} \exp \left[f_{\text{Edd}} \left(\frac{1 - \epsilon}{\epsilon} \right) \frac{\Delta t}{t_{\text{Edd}}} \right] B_{\text{merge}}, \quad (1)$$

where $t_{\text{Edd}} \simeq 0.45$ Gyr and $\Delta t = t(z_{\text{obs}}) - t(z_{\text{seed}})$. Cosmic time is evaluated with a flat Planck-style cosmology using $H_0 = 67.3 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.315$, and $\Omega_\Lambda = 0.685$.

The model is used in three directions: predicting a final mass, solving for the required lifetime-average f_{Edd} at fixed seed mass, and solving for the required seed mass at fixed accretion history. Implemented sensitivity dimensions include seed redshift, thin-disk radiative efficiency, Kerr spin, an illustrative photon-trapping efficiency above Eddington, and a multiplicative merger boost. Importantly, $B_{\text{merge}} = 2$ adds $\log_{10} 2 = 0.301$ dex to the mass; it does not double the exponential accretion rate.



Main-text prototype: points are source-reported v1 black-hole masses; curves are reference diagnostics, not fitted histories.
 All curves assume $z_{\text{seed}} = 30$, $\epsilon = 0.1$, and no merger boost; rankings are used for object triage.

Figure 1: The v1 JADES broad-line AGN sample in black-hole-mass-redshift space. The curves are selected reference growth histories, not fits to the observations. This overview establishes the timing problem before the model is inverted into required- f_{Edd} and required-seed diagnostics.

2.3 Stage 3 (v2): from scenario grids to observational triage

The first evaluation stage generated growth tracks, fixed-scenario compatibility summaries, and per-object maps. These products show where each measurement lies across seed mass, accretion, efficiency, merger, and formation-redshift space. They also revealed an important interpretive distinction: a fixed scenario can miss an object through *undergrowth* or *overgrowth*. Only the former directly signals insufficient growth; the latter can be resolved by less accretion, a smaller seed, or later formation.

The project therefore moved from generic compatibility scores to required-parameter rankings. The baseline ranking uses $z_{\text{seed}} = 30$, $\epsilon = 0.1$, and $B_{\text{merge}} = 1$, and keeps three concepts separate:

1. **Physical pressure:** required f_{Edd} for fixed seeds and required seed mass for fixed accretion histories.
2. **Measurement confidence:** robust/tentative status, missing fields, method caveats, and interpretation sensitivity.
3. **Follow-up value:** the potential for a better measurement to change the physical ranking.

This produces a measurement-level table rather than a single opaque score. Host-mass tension, growth pressure, confidence, and follow-up priority remain independently inspectable.

2.4 Stage 4: uncertainty propagation

Point estimates are insufficient near physically meaningful thresholds. The uncertainty layer draws 10,000 samples per measurement from a split-normal approximation to the reported asymmetric $\log_{10} M_{\text{BH}}$ uncertainty. For every draw, the pipeline recomputes the required f_{Edd} and required seed mass. It reports the 5th, 16th, 50th, 84th, and 95th percentiles, together with threshold probabilities.

Three systematic mass scenarios are evaluated separately:

$$\text{baseline,} \quad \log_{10} M_{\text{BH}} - 0.3 \text{ dex,} \quad \log_{10} M_{\text{BH}} + 0.3 \text{ dex.} \quad (2)$$

The statistical mass sampling is never blended across these systematic scenarios. This separation makes it possible to distinguish reported statistical uncertainty from an assumed method-scale shift.

2.5 Stage 5 (v3): source expansion and physical-object identity

The v3 release adds the complete 63-row Table 1 extraction from Taylor et al. (2025) [3] as a source-specific raw table, then applies $z \geq 4$ in the processing layer. The retained Taylor subset has 37 measurements of 36 physical objects, including 17 sources marked as little red dots (LRDs). LRD is stored as a phenotype; the object class remains broad-line AGN. The combined release therefore contains 60 measurements of 59 physical objects.

Every Taylor row preserves coordinates, redshift, total/narrow/broad $\text{H}\alpha$ fluxes and uncertainties, broad-line FWHM, formal virial-mass uncertainty, LRD and absorption-fit markers, field/program, and source caveats. The Reines et al. (2013) $\text{H}\alpha$ single-epoch calibration is tagged precisely, and its approximate ± 0.5 dex virial systematic is evaluated separately from formal errors and the common ± 0.3 dex comparison. Host mass, bolometric luminosity, and Eddington ratio remain unavailable rather than inferred from absent Table 1 values.

The measurement view retains both CEERS-2782 and RUBIES-EGS-50052. A separate crossmatch maps them to one stable physical-object ID; the higher-S/N RUBIES observation is the documented default for object-level ranking. No measurement is deleted or averaged. Summary tables are stratified by source, survey/field, LRD phenotype, and catalogue view, and mixed-selection rows are marked descriptive rather than demographic.

2.6 Stage 6 (v4): generalized identity and complementary WFSS samples

The v4 release adds all 20 Matthee et al. EIGER/FRESCO broad- $H\alpha$ emitters [4] and all 16 Lin et al. ASPIRE emitters [5] from authoritative arXiv source archives. Both papers use the Reines et al. (2013) $H\alpha$ single-epoch estimator and state a 0.5 dex calibration or intrinsic uncertainty. Native line luminosities, widths, photometry, continuum slopes, formal errors, LRD labels, absorption-fit flags, and row caveats are retained. Source-reported host mass and Eddington ratio remain missing; $H\alpha$ -derived bolometric luminosities are retained where published.

The identity layer performs a deterministic coordinate/redshift candidate search and refuses ambiguous matches. Review confirms one new cross-paper identity: Matthee GOODS-S-13971 is JADES GS-204851 at 0.02 arcsec separation and $\Delta z = 0.001$. The prior-release JADES measurement remains the default for longitudinal reproducibility, while the Matthee row remains independently rankable. Together with the Taylor pair, v4 contains 96 measurements but 94 physical objects.

The v4 science workflow preserves the baseline assumptions and reported asymmetric errors, keeps the common ± 0.3 dex comparisons, and adds separately named ± 0.5 dex Reines-calibration scenarios for Taylor, Matthee, and ASPIRE. Overall summaries explicitly mix unlike selection functions and are descriptive only.

3 Current scientific findings

The v2 v1-catalogue analysis identifies a stable high-leverage group, but the uncertainty analysis changes how strongly individual objects should be described. The v3 object-level ranking retains GN-38509 and GS-20057765 in the first two positions and adds RUBIES-EGS-49140 in third; the Taylor object is flagged for possible outflow contribution or an alternative non-AGN broad-line interpretation. Table 2 retains the detailed v1 probabilities that underlie the first comparison.

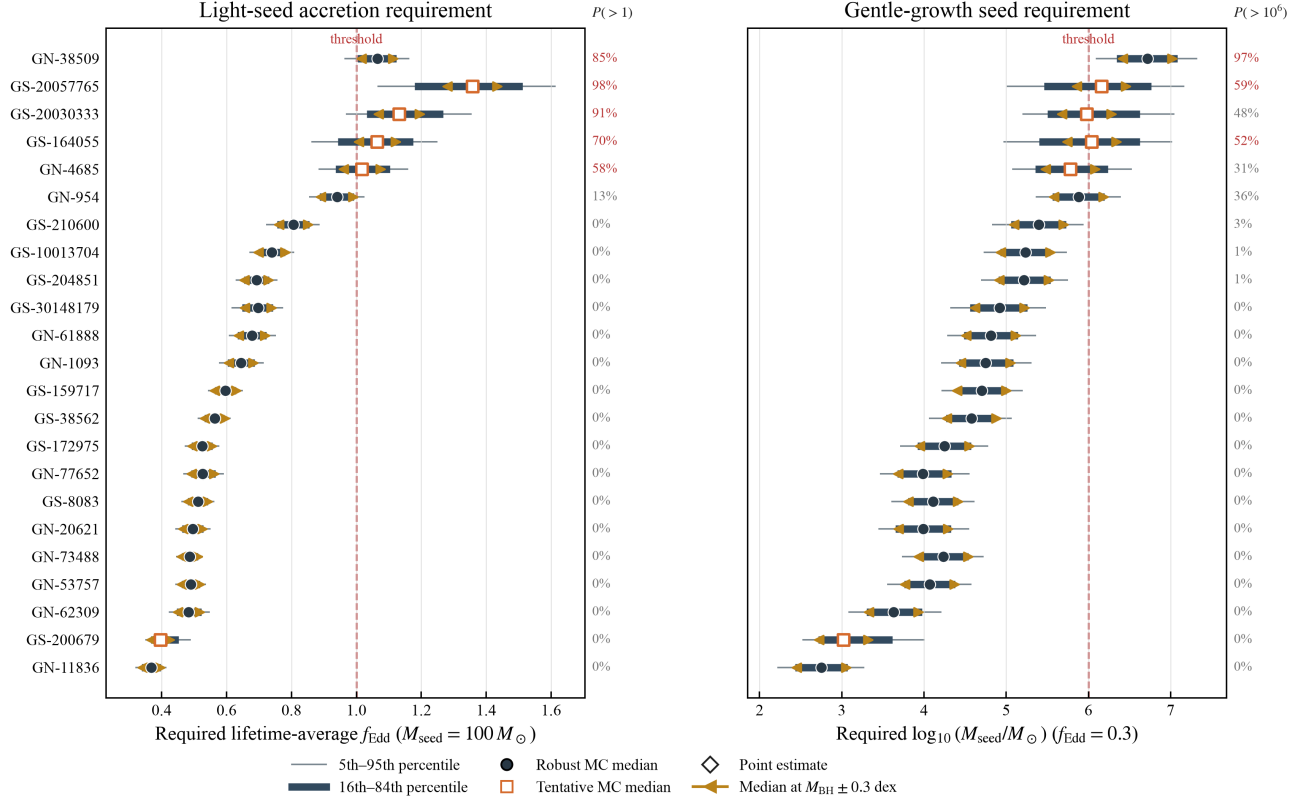
Table 2: Highest-leverage v1 measurements under the baseline uncertainty model.

Object	Quality	$P(f_{\text{Edd,req}}^{100} > 1)$	$P(M_{\text{seed,req}}^{0.3} > 10^6 M_{\odot})$	Status
GN-38509	robust	0.85	0.97	strongest robust pressure case
GS-20057765	tentative	0.98	0.59	strongest tentative follow-up target
GS-20030333	tentative	0.91	0.48	high leverage; elevated BH/host ratio
GS-164055	tentative	0.70	0.52	high leverage; extreme BH/host ratio
GN-4685	tentative	0.58	0.31	threshold and systematics sensitive
GN-954	robust	0.13	0.36	comparison/systematics object

GN-38509 is currently the clearest robust result: it remains demanding in both light-seed and gentle-growth diagnostics after propagating the reported mass uncertainty. GS-20057765 is more extreme in the light-seed calculation but is tentative and has a broader required-seed distribution. It is therefore a high-priority follow-up target rather than an equally secure theoretical constraint. GN-954 illustrates why probability statements matter: its point estimate is near a pressure boundary, but its baseline probability of requiring super-Eddington lifetime-average growth from a $100 M_{\odot}$ seed is only 0.13.

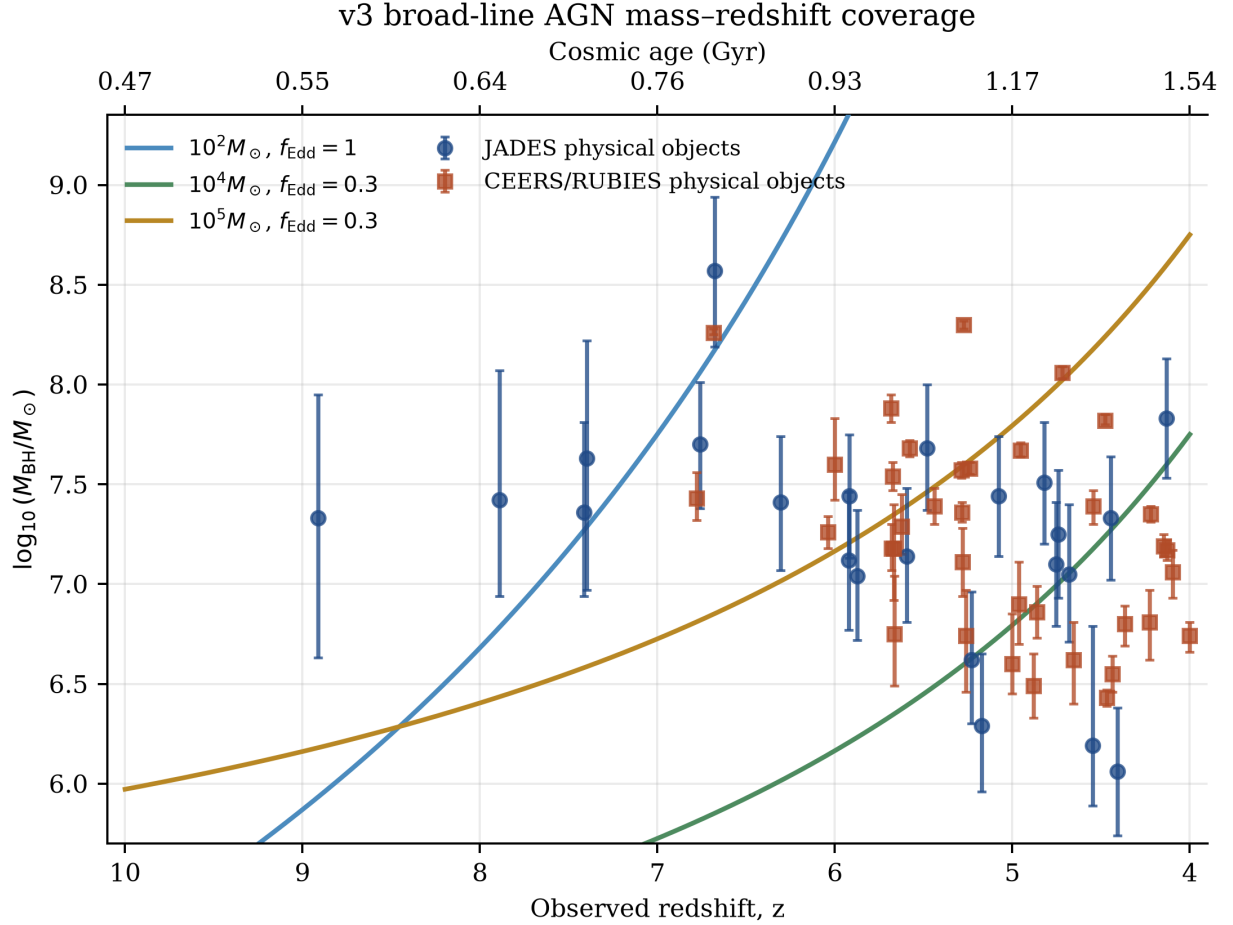
The current reported Eddington ratios are instantaneous observational estimates, whereas the model-derived f_{Edd} values are lifetime averages. Their difference is scientifically important but should not be interpreted as a contradiction until duty cycles and burst histories are modeled. A separate source-table cross-check identifies GN-11836 as internally inconsistent: its published $\log M_{\text{BH}} = 6.06$ and $\log L_{\text{bol}} = 44.11$ imply $\lambda_{\text{Edd}} \simeq 0.89$, while the table reports 0.11. All published values are retained pending clarification, and the measurement is downgraded to medium confidence for Eddington-ratio comparisons.

Uncertainty-aware growth-pressure ranking



Baseline Monte Carlo intervals propagate reported asymmetric M_{BH} errors; percentages give baseline threshold probabilities. Orange endpoints are separate $M_{\text{BH}} - 0.3$ and $+0.3$ dex systematic medians; all panels assume $z_{\text{seed}} = 30$, $e = 0.1$, and no merger boost.

Figure 2: Uncertainty-aware growth-pressure ranking. Thin and thick intervals show the baseline 5th–95th and 16th–84th percentiles. Circles and open squares denote robust and tentative Monte Carlo medians; diamonds are the original point estimates. Orange endpoints show the separate $M_{\text{BH}} \pm 0.3$ dex systematic medians. Red dashed lines mark $f_{\text{Edd}} = 1$ for a $100 M_{\odot}$ seed (left) and $M_{\text{seed}} = 10^6 M_{\odot}$ for $f_{\text{Edd}} = 0.3$ (right). Percentages give baseline probabilities of crossing the corresponding threshold.



Reference tracks assume $z_{\text{seed}} = 30$, $\epsilon = 0.1$, and merger boost 1; they are diagnostics, not fitted histories.

Figure 3: Frozen v3 physical-object mass-redshift coverage, stratified into JADES and Taylor CEERS/RUBIES selections. Reference tracks use the repository baseline and are diagnostic rather than fitted histories.

4 Release map and reproducibility

The project versions are chronological milestones: v1 is the pilot JADES catalogue and baseline evaluation, v2 is ranking/uncertainty/figure analysis on that frozen catalogue, v3 is the combined JADES + Taylor catalogue, and v4 generalizes identity while adding Matthee and ASPIRE. Source-paper versions, including arXiv suffixes, are independent provenance metadata. The canonical dependency map is:

1. `data/sources.md` and `data/raw/v1_raw.csv`: source provenance and extracted measurements.
2. `docs/catalogue-schema.md`: canonical fields, validation rules, and missing-value policy.
3. `src/standardize_data.py` and `scripts/process_data.py`: raw-to-processed catalogue construction.
4. `src/models.py`: cosmology and black-hole growth calculations.
5. `src/scoring.py` and `scripts/v1_evaluate.ipynb`: scenario evaluation and exploratory compatibility products.
6. `scripts/generate_v2_rankings.py`: v2 point-estimate physical-pressure and follow-up rankings from v1.
7. `scripts/generate_v2_uncertainty_rankings.py`: v2 deterministic Monte Carlo summaries and probability-based rankings.
8. `scripts/generate_v2_final_figures.py`: v2 publication-style figures driven by the v1-catalogue result tables.
9. `scripts/process_v3_blag.py` and `src/v3_catalogue.py`: v3 source combination, crossmatch, and object view.
10. `scripts/generate_v3_blag_science.py` and `src/v3_science.py`: v3 measurement/object evaluation, rankings, uncertainty, and summaries.
11. `scripts/generate_v3_final_figures.py`: frozen-v3 comparison figures.
12. `scripts/process_v4_blag.py`, `src/identity.py`, and `src/v4_catalogue.py`: v4 ingestion, reviewed matching, aliases, and measurement/object views.
13. `scripts/generate_v4_blag_science.py` and `src/v4_science.py`: v4 point and uncertainty products with source-specific systematics.
14. `tests/test_v1_v2_pipeline.py` and the `test_v3_*/test_v4_*` modules: model, schema, release, provenance, ranking, uncertainty, and regression checks.

The present test suite contains 70 passing checks. These protect implementation behavior, frozen release byte anchors, source counts and identity links, systematic separation, and current numerical rankings; they do not establish that the astrophysical assumptions are uniquely correct.

5 Limitations

The present atlas remains a selected-sample triage framework, and its conclusions should be read within these boundaries:

- v4 contains JADES, CEERS/RUBIES, EIGER/FRESCO, and ASPIRE selections, not a single complete population sample, so pooled rows cannot support demographic inference;

- the main growth diagnostics assume constant lifetime-average accretion rather than explicit duty cycles or bursts;
- the split-normal mass model approximates reported asymmetric uncertainties but is not a reconstruction of each source’s full posterior;
- the ± 0.3 dex cases are transparent comparison tests and the Taylor/Matthee/ASPIRE ± 0.5 dex cases are calibration sensitivities, not reconstructed systematic posteriors;
- the four stack-supported tentative broad- $H\beta$ candidates require individual spectroscopic confirmation before they can serve as secure theoretical constraints;
- the GN-11836 source-table Eddington-ratio inconsistency must be clarified before that ratio is used quantitatively;
- Taylor Table 1 does not publish host mass, bolometric luminosity, or Eddington ratio, so those diagnostics are unavailable rather than negative evidence; and
- deterministic physical-object linkage is explicit, but fixed thresholds and manual review are not a substitute for probabilistic crossmatching in crowded or low-precision samples.

These limitations motivate the next development stages rather than invalidating the release-stratified comparison framework.

6 Research roadmap

The next work should proceed in the following order.

1. Consolidate the manuscript and measurement-choice analysis

Integrate the completed v3 figures and verified v4 catalogue, then quantify how alternate literature measurements change rankings for the two multiply measured objects.

2. Add the next measurement-version layers

Attach Harikane and earlier discovery-paper measurements to stable v4 objects, improving alias coverage while retaining every literature row and rejecting ambiguous automatic matches.

3. Add object classes without erasing method differences

Extend the atlas to little red dots, X-ray candidates, lensed candidates, high-ionization-line candidates, and classical luminous quasars. Each class will retain distinct selection and mass-method metadata so that heterogeneous measurements are not treated as interchangeable.

4. Improve the uncertainty model

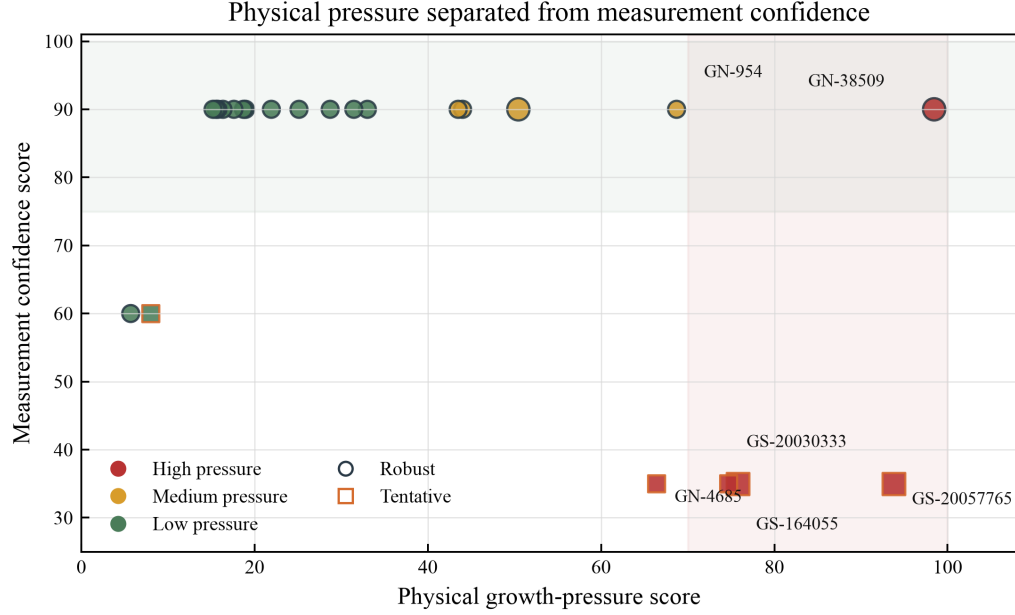
Propagate host-mass and lensing uncertainties where available, add method-specific black-hole-mass systematic scenarios, and use fuller posterior information when a source provides it. The output should continue to emphasize intervals and threshold probabilities rather than categorical point-estimate labels.

5. Replace constant-average growth with history diagnostics

Introduce duty-cycle and bursty-accretion models, compute the required duty cycle under fixed burst assumptions, and compare the reported current Eddington ratio with the required lifetime-average value without treating them as the same observable.

6. Convert the ranking into an observing strategy

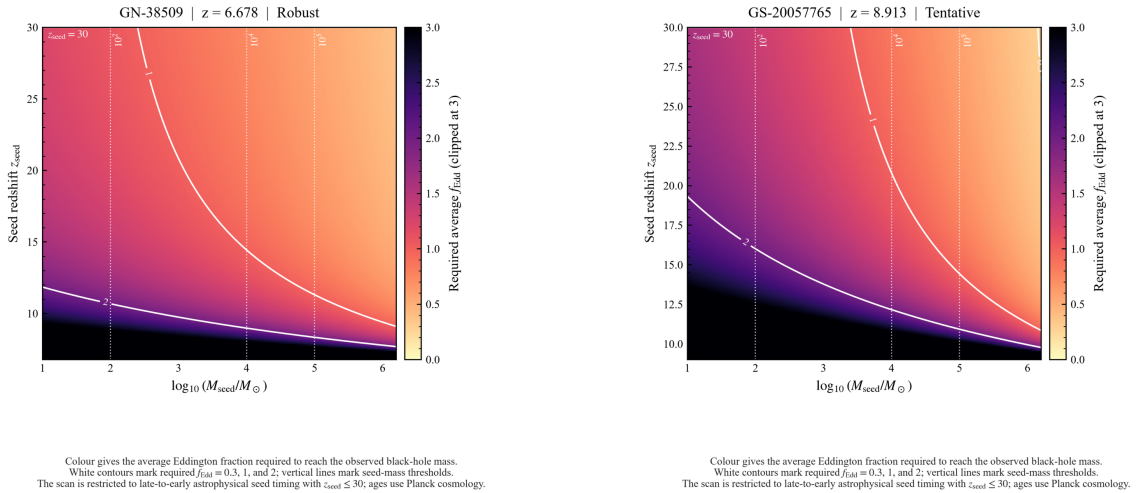
Produce a follow-up matrix specifying whether each object's interpretation is most sensitive to deeper spectroscopy, improved virial or dynamical mass constraints, host/AGN decomposition, X-ray observations, lensing reconstruction, or repeat measurements.



Triage view: physical growth pressure and measurement confidence are shown as separate axes. Large symbols mark extreme MBH/Mstar tension; tentative high-pressure objects are follow-up targets rather than firm formation claims.

Figure 4: Physical pressure and measurement confidence are intentionally separated. A tentative measurement may have high follow-up value because resolving it would strongly affect the interpretation, while a robust moderate-pressure measurement remains useful as a comparison anchor.

Spotlight seed-timing maps for the two strongest v1 leverage cases



Existing seed-redshift maps are reused as main-text spotlight prototypes. GN-38509: growth-pressure rank 1, $f_{\text{Edd}}(100M_{\odot}) = 1.06$; GS-20057765: growth-pressure rank 2, $f_{\text{Edd}}(100M_{\odot}) = 1.36$. Panels diagnose assumption sensitivity across seed mass and seed redshift; they do not select a unique formation path.

Figure 5: Seed-redshift spotlight maps for GN-38509 and GS-20057765. These maps show how the required lifetime-average accretion rate changes with seed mass and formation time. The full object gallery is retained for supplementary use; selected panels connect the sample ranking to object-level physical interpretation.

7 Conclusion

The completed v1–v4 work demonstrates the atlas concept: provenance-aware ingestion, validated standardization, explicit measurement/object identity, transparent physical modeling, point and Monte Carlo rankings, selection-aware summaries, release-specific visualization, and automated regression checks. v4 broadens the lower-redshift end of the same-class BLAGN comparison and demonstrates that a coordinate-level duplicate can be preserved without double-counting the physical object. This is not evidence that one formation pathway is required. The next phase should consolidate the manuscript, quantify measurement-choice sensitivity, and then add method-aware object classes and more realistic accretion histories.

References

- [1] Dayal, P. 2024, *A&A*, 690, A182.
- [2] Juodžbalis, I. et al. 2026, “JADES: comprehensive census of broad-line AGN from Reionization to Cosmic Noon revealed by JWST,” *MNRAS*, 546, stag086, arXiv:2504.03551.
- [3] Taylor, A. J. et al. 2025, “Broad-Line AGN at $3.5 < z < 6$: The Black Hole Mass Function and a Connection with Little Red Dots,” *ApJ*, 986, 165, doi:10.3847/1538-4357/add15b.
- [4] Matthee, J. et al. 2024, “Little Red Dots: an abundant population of faint AGN at $z \sim 5$ revealed by the EIGER and FRESCO JWST surveys,” *ApJ*, 963, 129, doi:10.3847/1538-4357/ad2345.
- [5] Lin, X. et al. 2024, “ASPIRE: Broad-line AGN at $z = 4$ –5 revealed by JWST/NIRCam WFSS,” *ApJ*, 974, 147, doi:10.3847/1538-4357/ad6565.